



Highly dynamic force control for experimental vibration analysis with an electrodynamic shaker

Christopher Lüke¹ · Ansgar Trächtler¹

Received: 17 May 2023 / Accepted: 13 August 2024 / Published online: 11 October 2024
© The Author(s) 2024

Abstract

A new control method aims at the precise high dynamic control of the force signal for experimental vibration analysis, which is generated by an electrodynamic shaker. A bending beam is used as a nonlinear test object. A design and a surrogate model of the test rig are shown and parameterized based on test rig measurements. The force control algorithm using input/output linearisation is described and implemented in Matlab/Simulink for simulative validation studies. Conclusions drawn from the mathematical description of the problem as well as simulation results show that the design of the contact between shaker and test object is crucial to achieve a high control bandwidth and at the same time reduce the energy consumption of the shaker. This leads to the practical application using a novel damping contact element. Finally, experimental test rig results are presented which show a closed loop bandwidth of at least 250 Hz for sinusoidal excitation signals.

Hochdynamische Kraftregelung für die experimentelle Schwingungsanalyse mit einem elektrodynamischen Shaker

Zusammenfassung

Eine neue Regelungsmethode zielt auf die präzise hochdynamische Regelung des Kraftsignals für die experimentelle Schwingungsanalyse, das von einem elektrodynamischen Schwingererzeuger erzeugt wird. Als nichtlineares Prüfobjekt wird ein Biegebalken verwendet. Ein Entwurfs- und ein Simulationsmodell des Prüfstands werden gezeigt und anhand von Prüfstandsmessungen parametrisiert. Der Algorithmus zur Kraftregelung unter Verwendung der Ein-/Ausgangslinearisierung wird beschrieben und in Matlab/Simulink für simulative Validierungsstudien implementiert. Die Schlussfolgerungen aus der mathematischen Beschreibung des Problems sowie die Simulationsergebnisse zeigen, dass die Gestaltung des Kontakts zwischen Schwingererzeuger und Prüfobjekt entscheidend ist, um eine hohe Regelbandbreite zu erreichen und gleichzeitig den Energieverbrauch des Schwingererzeugers zu reduzieren. Dies führt zur praktischen Anwendung eines neuartigen dämpfenden Kontaktelements. Abschließend werden experimentelle Prüfstandsergebnisse vorgestellt, die eine Regelbandbreite von mindestens 250 Hz für sinusförmige Anregungssignale zeigen.

1 Introduction

In the technical field, understanding the dynamic properties of components and structures is becoming increasingly important due to the growing prevalence of lightweight construction. In addition, model-based design and analysis represents an important part of the engineering activity. Precise and reliable measurement results are required for model validation. One way to obtain these results is to apply a defined force to the Structure Under Test (SUT). Challenges in this field are, among other things, high demands for the control dynamics in Hardware-in-the-loop tests while the dynamic of the SUT is unknown [10, 17].

Consent to participate The authors give consent to participate.
Consent for publication The authors give consent for publication.

✉ Christopher Lüke
christopher.lueke@hni.upb.de

Ansgar Trächtler
ansgar.traechtler@hni.upb.de

¹ Heinz Nixdorf Institute, Control Engineering and Mechatronics, Paderborn University, Fürstenallee 11, 33102 Paderborn, Deutschland

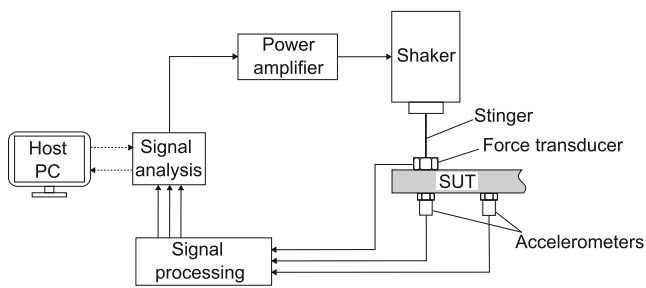


Fig. 1 Test set-up for experimental vibration analysis based on [5, 30]

Another area that requires a controlled force excitation is experimental vibration analysis, which is used here as an application to investigate the one-dimensional case. In the case of nonlinear structures, the challenges in determining measurement data are still subject of current research [22, 25]. Since these often exhibit amplitude dependence and respond with a spectrum of frequencies to a harmonic excitation, the correct measurement of individual modes is still a laborious and time-consuming procedure [11, 25].

In addition, the coupling between the shaker and the SUT is a source of error. The connection is made via a stinger, which does not transmit laterally acting forces and moments and thus ensures a unidirectional longitudinal force application to the structure as shown in Fig. 1. This can lead to distortions of the introduced force signal and displacement of resonance points [2, 13]. The influence of mass effects of force and acceleration sensing also plays a role [4, 12, 21]. Another challenge is the force dropout phenomenon, where the in-phase oscillation of the shaker and SUT causes a dropout of the force signal near the SUT resonances [14, 18, 29, 30].

The above suggests that the force signal applied to the structure should be subject to a closed-loop control. By controlling it, a known and constant amplitude level could be ensured over the entire frequency range to be measured. In addition, the avoidance of non-harmonic components in the excitation signal would be possible. Therefore one finds demands for the regulation of the excitation signal, for example [1, 19, 28].

In the field of environmental simulations or load tests, controllers are often used to apply the previously recorded load data as accurately and reproducibly as possible to the structure to be measured [15, 29, 30]. However, since noise-like, broadband signals are involved, these methods are not suitable for the application considered here, which is based on comparably narrow-banded sinusoidal excitations.

Vibration controllers are often used for vibration analysis on complex and cost-intensive mechanical structures (e.g. satellites) to ensure a non-destructive test sequence. Reference sensors are attached to the SUT at critical points; the controller ensures that a specified load profile is maintained at these points or that certain load limits are not exceeded

[15]. Input notching is often performed in this context [7]. In this case, the excitation amplitude is lowered in the range of resonances in order not to overload the SUT. However, due to the amplitude dependence of the nonlinear behaviour of mechanical structures, this point has to be considered critically with respect to their exact dynamic description.

In [9], a method is proposed in which higher-harmonic components in the excitation signal are compensated so that a mono frequency (harmonic) setpoint results. For this purpose, the higher-harmonic components are identified in an iterative process and compensated by an inverse feed-forward to the reference signal. The disadvantages of this method lie in the complex and lengthy calculation, which must be performed in real time. In addition, the method is poorly conditioned, so that numerical instabilities frequently occur.

The company LMS installed a control system in its vibration analysis systems, which, however, does not represent a closed-loop control in the classical sense. Before the actual measurement, a preliminary identification is carried out using a noise signal and thus creating a system frequency response function for the excitation system. On this basis, the target amplitudes for the actual measurement are calculated. During this measurement, the excitation force curves calculated based on the system identification are compared with the actual ones. In case of deviations, the nominal signal is adjusted stepwise in an iteration loop [24, 32]. This procedure is very time-consuming, especially for nonlinear mechanical structures, since in these cases the pre-identified behaviour sometimes deviates significantly from the actual behaviour in the measurement.

All this shows that there is a demand for force control in experimental vibration testing, while comprehensible solutions based on feedback control are still lacking. Further approaches for the control of force signals come from robotics. However, these either assume quasi-static loads in unknown environments [17, 31] or require the dynamics of the test object to be known. An interesting concept here is the use of a so-called “compliance spring”, which is a series elastic connection between the excitation unit and the test structure. This decouples the actuator from the test structure and eliminates the need to model the SUT. Similar thoughts are also expressed in [23, 27]. While stiffness is favourable for precision and stability of position control, increased series elasticity makes stable force control easier to achieve [20]. Force control is also required when performing earthquake tests with large hydraulic actuators and in [3], a compliance spring is successfully used.

However, this concept has not yet been applied to high-frequency excitations. Nevertheless, due to the advantages mentioned above, this concept is here adapted to the application of experimental vibration analysis. Due to the low-pass characteristics of the compliance spring, a damping

element will be added to achieve a high control bandwidth while lowering the necessary amount of control energy. The overall goal of the application is to control a sinusoidal force with varying frequencies and a constant amplitude, at least apart from the resonances of the SUT, whose dynamics are assumed to be unknown to the control algorithm. There shall be no need for a priory identification of the SUT and iterative parameter adaptation during the testing process.

This paper is structured as follows: Sect. 2 describes the design model on which the concept for force control presented in Section 3 is based. In Sect. 4, measured data is used to extend the design model to a surrogate model, based on which the control is to be tested simulatively. In addition, a state observer is necessary to realise the control. This is described in Sect. 5. The interaction of all subsystems is investigated simulatively in Sect. 6, as well as the influence of damping parameters on the force control loop. Finally, the implementation of the control concept on the test bench as well as measurement data of the control system are shown. The last Section gives a summary and an outlook.

2 Design model

The basic model of the experimental setup is shown in Fig. 2. The electrodynamic shaker can be divided into an electrical and a mechanical subsystem, which are coupled to each other via the Lorentz force and the motion-induced counter voltage, respectively [26]. The mechanical subsystem is in turn coupled to the SUT. This is classically done by means of a stinger and a load cell, which can alternatively be modelled by a contact stiffness.

The following applies to the actuator, an oscillating armature with nonlinear spring behaviour:

$$m_A \ddot{x}_A = F_A - F_C - d_A \dot{x}_A - c_A x_A - c_{A2} x_A^2 - c_{A3} x_A^3, \quad (1)$$

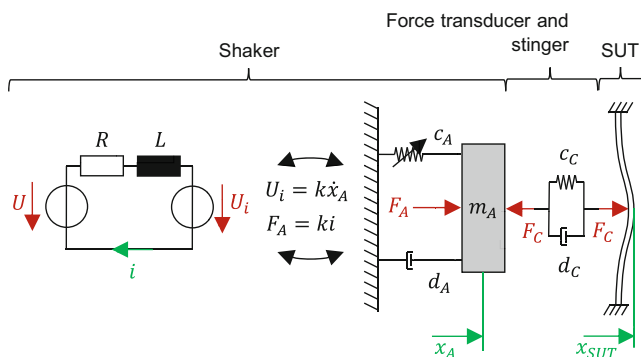


Fig. 2 Design model of the schematic experimental setup

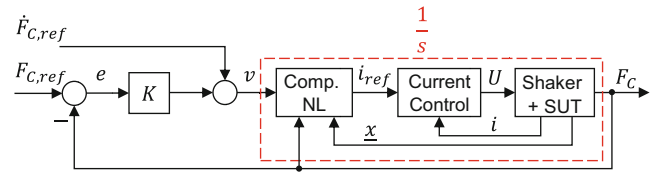


Fig. 3 Contact force control scheme with compensated nonlinearity, current control and force feedback

with $F_A = k \cdot i$, where the contact force is modelled after the common model of a linear spring-damper element:

$$F_C = d_C (\dot{x}_A - \dot{x}_{SUT}) + c_C (x_A - x_{SUT}). \quad (2)$$

Since the properties of the SUT are to be considered unknown, it is taken as a measurable disturbance for the actuator. Thus, the mechanical subsystem in the state space can be described as follows:

$$\begin{aligned} \begin{bmatrix} \dot{x}_A \\ \ddot{x}_A \end{bmatrix} &= \underline{a}(\underline{x}) + \underline{b}(\underline{x}) u + \underline{e}(\underline{z}) \\ &= \begin{bmatrix} \dot{x}_A \\ -\frac{1}{m_A} ((d_C + d_A) \dot{x}_A - (c_C + c_A) x_A - c_{A2} x_A^2 - c_{A3} x_A^3) \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 \\ \frac{1}{m_A} \end{bmatrix} F_A + \begin{bmatrix} 0 \\ \frac{1}{m_A} (c_C x_{SUT} + d_C \dot{x}_{SUT}) \end{bmatrix}, \end{aligned}$$

$$\begin{aligned} y = F_C &= \underline{c}(\underline{x}) + \underline{f}(\underline{z}) \\ &= [c_C x_A + d_C \dot{x}_A] + [-c_C x_{SUT} - d_C \dot{x}_{SUT}]. \end{aligned} \quad (3)$$

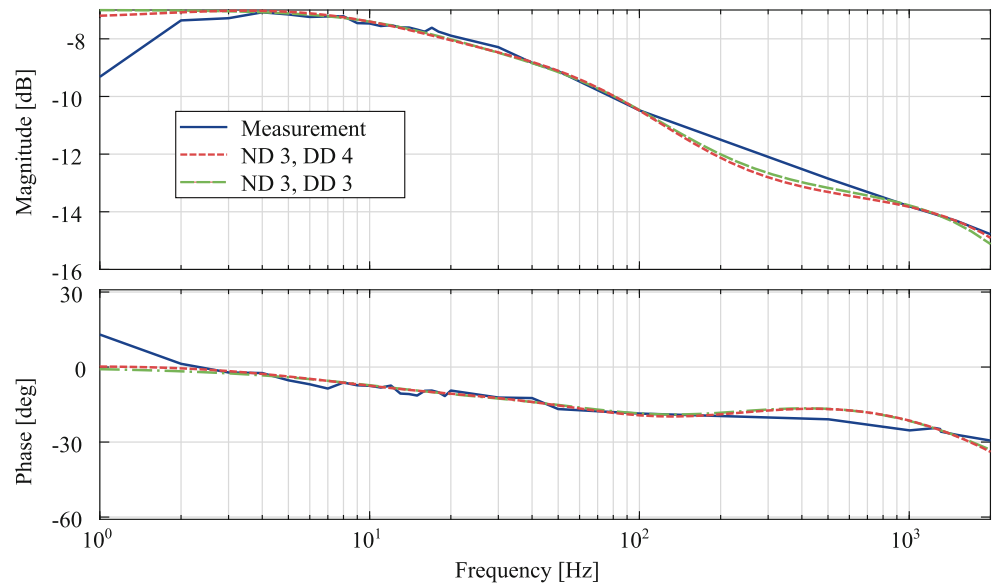
It can be seen that the disturbance variables $\underline{z} = [x_{SUT} \dot{x}_{SUT}]^T$ occur in the controlled variable F_C as direct feed through. This problem is addressed in the control concept in Sect. 3. The modelling of the electrical subsystem is discussed in Sect. 4.

3 Force control concept

The aim of the force control is to introduce a harmonic sinusoidal contact force F_C into the SUT. The dynamic behaviour of the SUT is assumed to be unknown. Only the motion quantities of the SUT are available by measurement. Thus, the SUT must be considered as a disturbance for the excitation unit, as shown before. As a result, the disturbance variables x_{SUT} and $\dot{x}_{SUT}$ occur in the controlled variable F_C . This prevents the use of control concepts which are based on a two-degree-of-freedom structure with feed-forward control and feedback of the system's states.

The controller design for the contact force is carried out using the method of input/output linearisation, which has already proven to be useful in the context of the force control of hydraulic excitation units [10]. The actuator force F_A in the mechanical subsystem serves as input. It is assumed

Fig. 4 Measured and identified frequency responses of the transfer behaviour from control voltage to coil current with the armature fixed



that the actuator force is set sufficiently fast by the underlying current control, which is validated in Sect. 4. Thus, the consideration of the motion quantities of the SUT in the linearisation is possible. For the calculation, the time derivative of the contact force is first formed from Eq. 3:

$$\begin{aligned} \dot{y} = \dot{F}_C &= \frac{\partial c(\underline{x})}{\partial \underline{x}} \dot{\underline{x}} + \frac{\partial f(\underline{z})}{\partial \underline{z}} \dot{\underline{z}} \\ &= \frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{a}(\underline{x}) + \frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{b}(\underline{x}) u + \frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{e}(\underline{z}) + \frac{\partial f(\underline{z})}{\partial \underline{z}} \dot{\underline{z}}. \end{aligned} \quad (4)$$

In the case of the system described in Eq. 3, $\frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{b}(\underline{x}) = \frac{d_C}{m_A} \neq 0$, which gives us the actuator force necessary for linearisation.

$$\begin{aligned} F_A &= \left(\frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{b}(\underline{x}) \right)^{-1} \\ &\cdot \left[v - \frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{a}(\underline{x}) - \frac{\partial c(\underline{x})}{\partial \underline{x}} \underline{e}(\underline{z}) - \frac{\partial f(\underline{z})}{\partial \underline{z}} \dot{\underline{z}} \right] \quad (5) \\ &= F_C + \frac{m_A}{d_C} (v + c_C (\dot{x}_{\text{SUT}} - \dot{x}_A) + d_C \ddot{x}_{\text{SUT}}) + d_A \dot{x}_A \\ &\quad + c_A x_A + c_{A2} x_A^2 + c_{A3} x_A^3. \end{aligned}$$

The actuator force calculated from Eq. 5 does not contain any model knowledge of the SUT dynamics, while only requiring real-time measurements of the SUT response. It is realised by specifying a reference actuator force $F_{A,\text{ref}}$ which is converted via the coupling factor k and fed to the internal current control cascade. This results in a simple

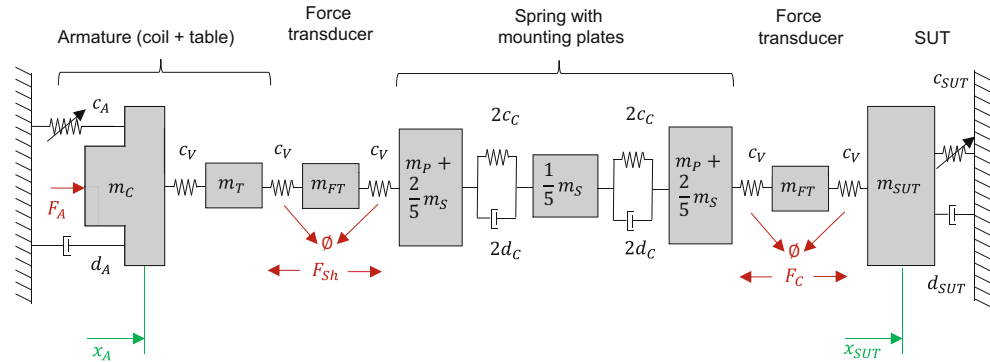
integrator behaviour from the new linear input $\dot{F}_C = v$ to the contact force F_C . The controller approach for the new linear input is then

$$v = \dot{F}_{C,\text{ref}} + K (F_{C,\text{ref}} - F_C), \quad (6)$$

resulting in exponentially stable error dynamics for the force control loop which is displayed in Fig. 3. In the ideal case, the controller gain K corresponds directly to the bandwidth of the force control.

From the calculation rule for F_A (Eq. 5), conclusions can be drawn directly as to how the parameters of the mechanical subsystem affect the control. Since the damping parameter d_C of the Shaker-SUT contact is in the denominator, it should be appropriately large to keep the required Lorentz force low. Conversely, for the contact stiffness c_C , it should not be too large because it amplifies the deviation $\dot{x}_{\text{SUT}} - \dot{x}_A$. Moreover, a large value of c_C leads to high system eigenvalues, which cause problems in observer design. In turn, too low a contact stiffness leads to a reduced control bandwidth due to the stronger low-pass behaviour of the spring and power limitations of the shaker. Furthermore, a low damping d_A of the armature guidance seems desirable for a low actuating energy. These considerations will be checked simulatively in Sect. 6. To be able to implement the control concept in practice, it is therefore necessary to use a precisely known and suitably parameterized contact element, which supplements the conventional stinger setup in the experimental vibration analysis.

Fig. 5 Surrogate model of the mechanical subsystem with bending beam as SUT



4 Surrogate model

The design model from Sect. 2 assumes a massless spring as a surrogate model for the contact stiffness. Measurements on the real system will be used to investigate the extent to which the mass of the real spring has an influence on the dynamics of the force transmission. Based on the measurements, a surrogate model is developed and used for the simulative investigation of the control concept.

4.1 Electrical subsystem

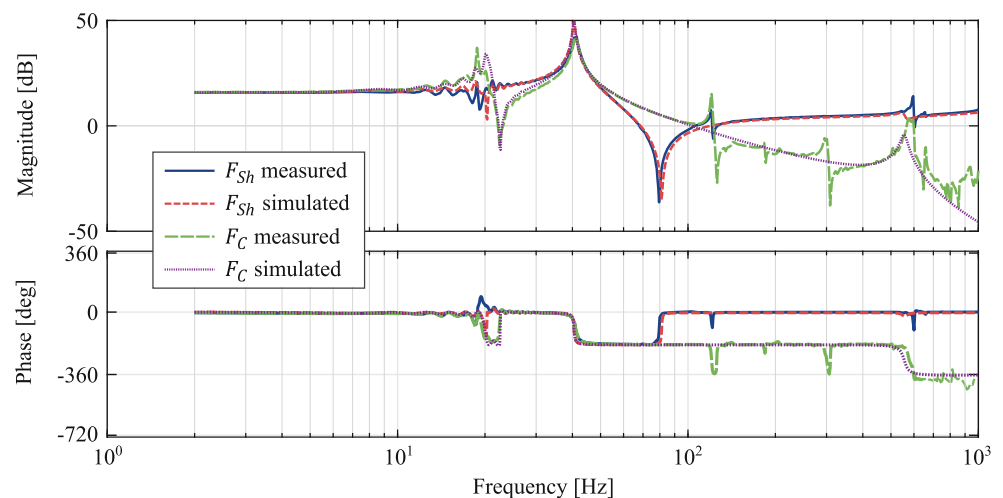
The tests were carried out using a TIRA shaker, model TIRAvib 52122-M. This shaker has a nominal force of 200 N and a vibration displacement p - p of 25.4 mm and can be used over a frequency range of 2 to 2000 Hz. A tension-compression steel spring was used to couple the shaker to a mild steel bending beam (length: 750 mm, width: 25 mm, thickness: 3 mm) clamped on both sides. The beam parameters were designed simulatively with the aim of achieving Duffing behaviour with multiple response frequencies in a frequency range above the shaker resonance, while at the same time respecting the actuator and measurement limitations. A 208C02 piezo force sensor from PCB Piezotronics was mounted on both the shaker and beam ends of the

spring. Slim steel plates were welded to the spring on both ends to accommodate the required threaded holes.

The system is controlled using real-time hardware from dSPACE (DS1103 board). In addition, two 8702B acceleration sensors from Kistler, an LD1605-10 laser displacement sensor from Micro-Epsilon, and a PR30 current clamp from LEM were used. The measured forces are denoted as F_{Sh} and F_C , respectively. The acceleration of the armature $\ddot{x}_A$ was measured at the mandrel, which guides the movement of the armature through a slide bearing at the lower end of the shaker body. The movement of the beam at the initiation point of the force was determined by measuring x_{SUT} und $\ddot{x}_{SUT}$.

First, tests were carried out for the electrical subsystem with both fixed and freely oscillating armature to identify the resistances, inductances and coupling factor between the electrical and mechanical subsystems. Figure 4 shows the measured transfer behaviour from control voltage to coil current with the armature fixed. Thus, there is no counter voltage reducing the control voltage. It can be seen that this is not a standard PT1 transfer function. Therefore, with the aid of optimisation in Matlab, two transfer functions with a numerator degree (N) of three and denominator degrees (D) of three and four, respectively, were identified which sufficiently accurately represent the behaviour in the

Fig. 6 Measured and simulated frequency responses of coil current to the measured forces with a pure steel spring as contact element



frequency range up to the shaker's upper operating frequency of 2 kHz:

$$G_{N3D3}(s) = \frac{0.03s^3 + 3891s^2 + 4.47 \cdot 10^6s + 5.25 \cdot 10^8}{s^3 + 1.79 \cdot 10^4s^2 + 1.23 \cdot 10^7s + 1.13 \cdot 10^9},$$

$$G_{N3D4}(s) = \frac{4288s^3 + 4.87 \cdot 10^6s^2 + 5 \cdot 10^8s + 5.47 \cdot 10^9}{s^4 + 2.1 \cdot 10^4s^3 + 1.34 \cdot 10^7s^2 + 1.14 \cdot 10^9s + 1.26 \cdot 10^{10}}.$$

The transfer functions are later used for the current control and the observer model, respectively.

4.2 Mechanical subsystem

Regarding the mechanical subsystem, [8] states that a consideration of the spring mass is necessary for a correct modelling of the force transmission. Furthermore, the design model does not consider the screw connection between coil and table [26]. Thus, it will be extended in the following. All masses of the table, the force sensors and the mounting plates of the spring are vibrationally connected via screw stiffnesses as shown in Fig. 5. The spring mass is divided 2/5 to each of the plates and 1/5 is modelled as a separate spring mass. The measured forces are calculated as the average of the forces from the bolted connections on both sides of the sensors. The beam can alternatively be modelled by a spring-damper-mass oscillator. The cubic part in the spring stiffness results in the following SUT dynamics:

$$m_{SUT}\ddot{x}_{SUT} + d_{SUT}\dot{x}_{SUT} + c_{SUT1}x_{SUT} + c_{SUT3}x_{SUT}^3 = F_C. \quad (7)$$

This model of the SUT will be used only for the simulation-based validation of the controller design. For the force control scheme, it is assumed to be unknown.

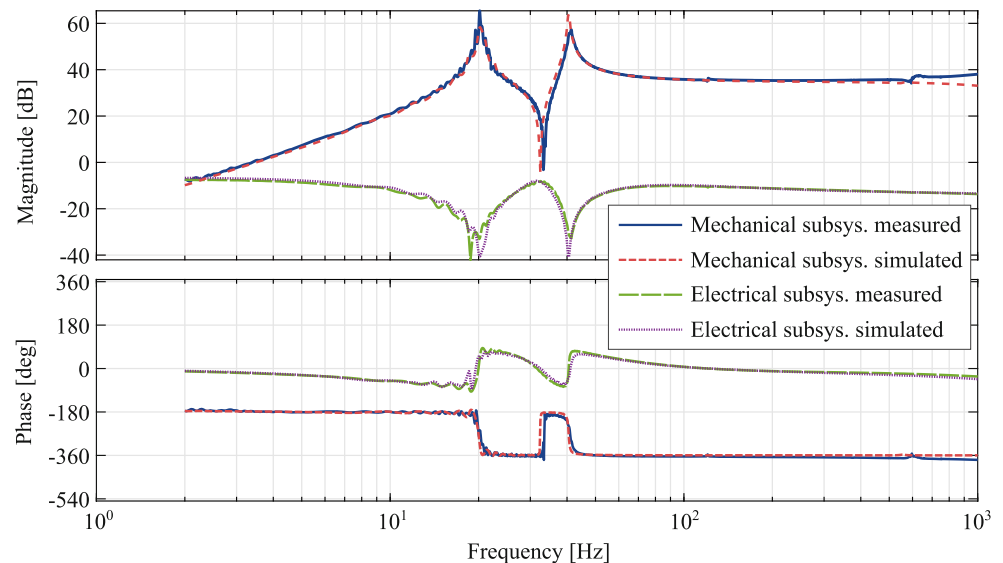
Table 1 Parameters of the mechanical subsystem

Parameter	Description	Value
c_A	Armature stiffness (linear component)	3920 N/m
c_{A2}	Armature stiffness (quadratic component)	5495 N/m ²
c_{A3}	Armature stiffness (cubic component)	$7.93 \cdot 10^6$ N/m ³
c_C	Contact spring stiffness	7600 N/m
c_{SUT}	Beam stiffness (linear component)	6203 N/m
c_{SUT3}	Beam stiffness (cubic component)	$8.56 \cdot 10^7$ N/m ³
c_V	Screw connections stiffness	$2.63 \cdot 10^7$ N/m
d_A	Armature damping (ball bearing)	1.5 Ns/m
d_C	Contact spring damping	1 Ns/m
d_{SUT}	Beam damping	2.19 Ns/m
m_S	Contact spring mass	6.63 g
m_{FT}	Force transducer mass	27 g
m_P	Mounting plate mass	13.68 g
m_C	Coil mass	155 g
m_{SUT}	Modal mass of the bending beam in the 1st natural frequency	174 g
m_T	Table mass	67 g

The shaker was now excited with a voltage sweep from 2 to 1000 Hz and an amplitude of 2 V. Figure 6 shows the frequency responses of coil current (or Lorentz force) to the measured force signals (green and red) using a pure steel spring as the contact element.

It can be seen that several resonances occur. The resonance at 20 Hz can be attributed to the actuator mass $m_A = m_C + m_T$ and the spring c_A . The antiresonance at 24 Hz in the signal of F_C occurs at the resonant frequency

Fig. 7 Measured and simulated frequency responses of current to shaker acceleration and input voltage to current, respectively



of the SUT, which in Fig. 18 is shown to not depend on the excitation amplitude as one would expect from the cubic stiffness term. In shaker testing, this effect is referred to as the force dropout phenomenon. From a control theoretical point of view, this means that an invariant transmission zero exists in the transfer behaviour of the control variable, which cannot be compensated for by the force control algorithm. The resonance point at 42 Hz can be attributed to the armature and the spring stiffnesses c_A and c_C . At 126 Hz, another resonance point occurs, which is at the third eigenmode of the flexural beam. Note that the Duffing Eq. 7 does not include the higher beam modes, so these are not included in the simulated frequency responses. In the signal of F_{Sh} , an antiresonance at 85 Hz can also be seen. Thus, there is a significant deviation of the forces on both ends of the contact spring.

The equations of the model were implemented in Matlab/Simulink. For simulation, the model was excited with the same signal as the test rig. The model parameters were determined by nonlinear least squares optimisation, minimising the error between the simulated and measured signals of the forces F_{Sh} and F_C . The result of this investigation is that the surrogate model reproduces the measured data sufficiently accurately. All resonances and antiresonances are reproduced and are also subject to only minor deviations in terms of quantity. The parameters of the model are shown in Table 1.

The model also reproduces the coupling of the electrical and mechanical subsystems well. Figure 7 shows the frequency responses of the mechanical (coil current to shaker acceleration) and electrical (input voltage to coil current) subsystems of the shaker. The resonance points of the mechanical subsystem coincide with the resonance points of the forces. At the same time, dips in the current waveform can be found here, which are caused by the back EMF.

4.3 Current control

Finally, the assumption that the current control can be regarded as sufficiently fast for the superimposed force control should be checked at this point. It is shown schemati-

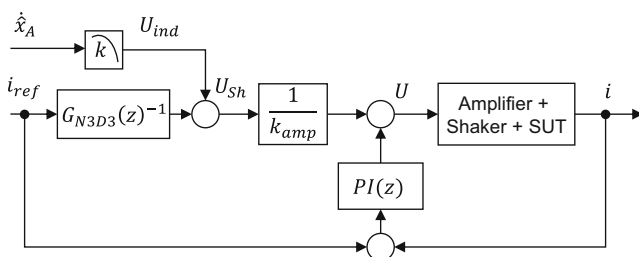


Fig. 8 Scheme of current control with feedforward and induced reverse voltage compensation

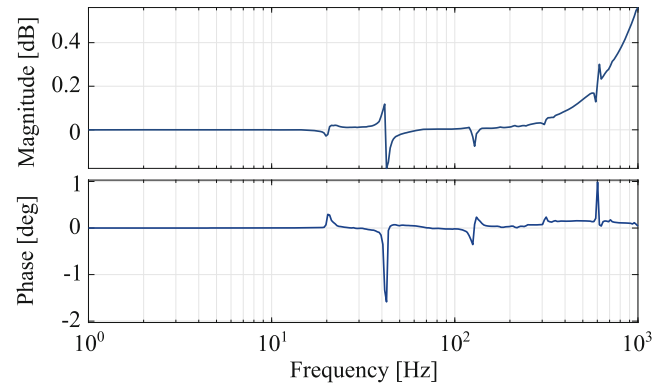


Fig. 9 Measured frequency response of the current control loop with the beam coupled

cally in Fig. 8 and is designed as a two-degree-of-freedom structure.

The identified transfer function with equal numerator and denominator degrees from Fig. 4, $G_{N3D3}(s)$, is discretised and inverted for usage as feedforward. Furthermore, by observing the armature speed $\dot{x}_A$, the induced reverse voltage U_{ind} is compensated. The control voltage U of the real-time hardware is obtained after converting the voltage U_{Sh} applied to the shaker via the power amplifier's gain factor k_{amp} , which has been shown to be linear in the range of the measurements depicted here. To this is added the intervention of the discrete PI controller, which has the controller gains $K_P = 0.3$ and $K_I = 7000$.

Figure 9 shows the measured frequency response of the closed current control loop. The nominal signal is a current sweep from 1 Hz to 1 kHz with 0.15 A amplitude. The sample time of the real-time board is 67 μ s (15 kHz). The shaker is coupled with a bending beam clamped on both sides as SUT. It can be seen that the bandwidth of the control is at least 1 kHz. The mechanical resonances of the shaker and the beam cannot be fully compensated, but they have an insignificant influence on the control performance. The assumption for the force control concept that the Lorentz force can be set sufficiently fast is therefore valid.

5 State observer design

The use of an observer has two objectives in the context of force control. On the one hand, the non-measurable state variables x_A and $\dot{x}_A$ are to be estimated. On the other hand, the measurement signals are to be freed from noise as far as possible in order to generate a control voltage U that is as smooth as possible. This is important because noise is amplified both by the controller gain K and by the calculation of F_A according to Eq. 5 and can easily drive the control signal into the limit.

The Kalman filter is chosen as the observation method. This has proven itself for the state estimation of mechanical systems and explicitly considers measurement and process noise. Moreover, it can be implemented quickly in Matlab/Simulink. With sufficient accuracy, a linear behaviour of the system can be assumed over large parts of the operating range as the deflections of shaker and beam correspond to the linear part of their stiffness curves. In general, the state equations of the linear Kalman filter are as follows:

$$\hat{\underline{x}}_K = \underline{A}_K \hat{\underline{x}}_K + \underline{B}_K \underline{u}_K + \underline{L} \left(\underline{y}_K - \underline{C}_K \hat{\underline{x}}_K - \underline{D}_K \underline{u}_K \right). \quad (8)$$

The general state equations of the underlying model, which are detailed below, are

$$\begin{aligned} \underline{x}_K &= \underline{A}_K \underline{x}_K + \underline{B}_K \underline{u}_K + \underline{G} \underline{w}_p, \\ \underline{y}_K &= \underline{C}_K \underline{x}_K + \underline{D}_K \underline{u}_K + \underline{H} \underline{w}_p + \underline{w}_m. \end{aligned} \quad (9)$$

For the white noise signals of the process $\underline{w}_p$ and the measurement $\underline{w}_m$ respectively holds

$$\begin{aligned} \underline{E}(\underline{w}_p) &= \underline{E}(\underline{w}_m) = \underline{0}, \quad \underline{E}(\underline{w}_p \underline{w}_p^T) = \underline{Q}, \\ \underline{E}(\underline{w}_m \underline{w}_m^T) &= \underline{R}, \quad \underline{E}(\underline{w}_p \underline{w}_m^T) = \underline{N}, \end{aligned} \quad (10)$$

with $\underline{Q}$, $\underline{R}$ and $\underline{N}$ as noise covariance matrices, which must be chosen accordingly for the calculation of the filter matrix $\underline{L}$.

Considering the high number of states and the large stiffnesses in the surrogate model, it is not suitable for a real-time observable design. However, no simplified model was found that satisfactorily reconstructs simultaneously all

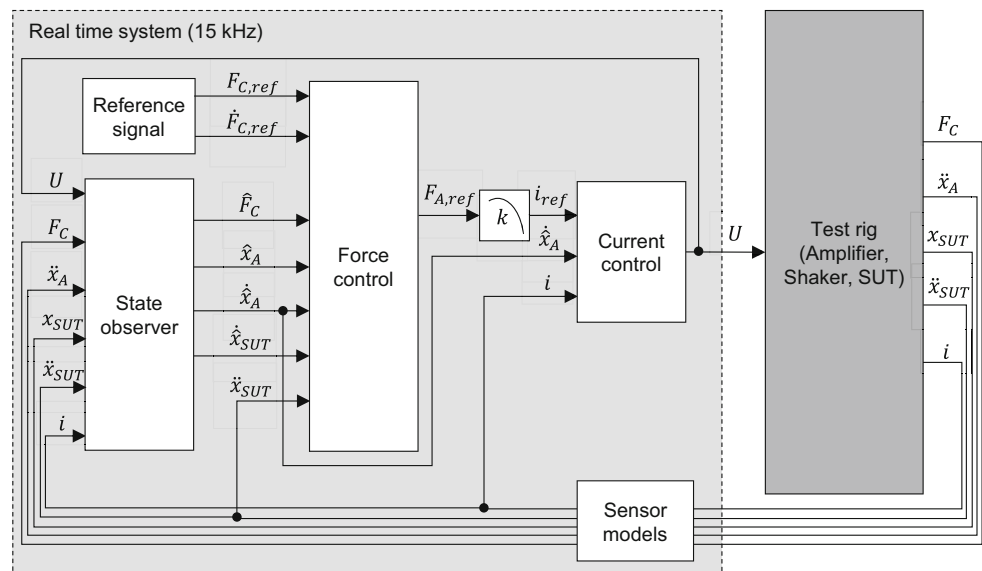
three observable quantities F_C , x_A and $\dot{x}_A$. Instead, two parallel observers are used. The first observer is used to smooth the force signal and is based on the linearised simple design model (Eq. 3).

$$\begin{aligned} &\begin{bmatrix} \dot{\hat{x}}_A \\ \dot{\hat{x}}_A \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ \frac{-c_C - c_A}{\hat{m}_A} & \frac{-d_C - d_A}{\hat{m}_A} \end{bmatrix} \begin{bmatrix} \hat{x}_A \\ \dot{\hat{x}}_A \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ \frac{k}{\hat{m}_A} & \frac{c_C}{\hat{m}_A} & \frac{d_C}{\hat{m}_A} \end{bmatrix} \begin{bmatrix} i \\ x_{SUT} \\ \dot{x}_{SUT} \end{bmatrix}, \\ &\hat{y} = \hat{F}_C \\ &= \begin{bmatrix} c_C & d_C \end{bmatrix} \begin{bmatrix} \hat{x}_A \\ \dot{\hat{x}}_A \end{bmatrix} + \begin{bmatrix} 0 & -c_C & -d_C \end{bmatrix} \begin{bmatrix} i \\ x_{SUT} \\ \dot{x}_{SUT} \end{bmatrix}. \end{aligned} \quad (11)$$

The input variables i and x_{SUT} are measured, while $\dot{x}_{SUT}$ is derived phase-free from x_{SUT} and $\ddot{x}_{SUT}$. The equivalent mass $\hat{m}_A$ represents the oscillating mass of the shaker and considers not only the armature mass but also the force sensor and parts of the contact element. By suitable choice of the covariance matrices, a high observer quality can be achieved despite the model uncertainty: $\underline{Q}_F = \text{diag}[5 \cdot 10^{44} \cdot 10^5]$, $\underline{R}_F = 4 \cdot 10^{13}$, $\underline{N}_F = \underline{0}$.

The second observer is used to estimate the actuator states. The control voltage U and the measured shaker-side force F_{sh} are used as inputs, which removes the influence of the contact element and the SUT from the equations:

Fig. 10 Schematic structure of the simulation model



$$\begin{aligned}
 & \begin{bmatrix} \hat{\ddot{x}}_A \\ \hat{\ddot{x}}_A \\ \hat{i}_1 \\ \hat{i}_2 \\ \hat{i}_3 \\ \hat{i}_4 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{-c_A}{\hat{m}_A} & \frac{-d_A}{\hat{m}_A} & \frac{k}{\hat{m}_A} \underline{C}_{N3D4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -k \underline{B}_{N3D4} & \underline{A}_{N3D4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_A \\ \hat{\dot{x}}_A \\ \hat{i}_1 \\ \hat{i}_2 \\ \hat{i}_3 \\ \hat{i}_4 \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{\hat{m}_A} \\ 0 & 0 \\ k_{\text{amp}} \underline{B}_{N3D4} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U \\ F_{Sh} \end{bmatrix}, \\
 &\underline{y} = \begin{bmatrix} \hat{i} \\ \hat{\ddot{x}}_A \end{bmatrix} = \begin{bmatrix} 0 & 0 & \underline{C}_{N3D4} \\ \frac{-c_A}{\hat{m}_A} & \frac{-d_A}{\hat{m}_A} & \frac{k}{\hat{m}_A} \underline{C}_{N3D4} \end{bmatrix} \hat{\underline{x}} \\
 &+ \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{\hat{m}_A} \end{bmatrix} \begin{bmatrix} U \\ F_{Sh} \end{bmatrix}. \quad (12)
 \end{aligned}$$

For the electrical subsystem, the identified transfer function $G_{N3D4}(s)$ is converted into a state space model with four states $i_k, k = 1 \dots 4$. This eliminates the feedthrough, which was advantageous for feedforward control of the current. The estimated current $\hat{i}$ and acceleration of the armature $\hat{\ddot{x}}_A$ serve as output variables for the filter. To achieve the best possible noise suppression, the covariance matrices of

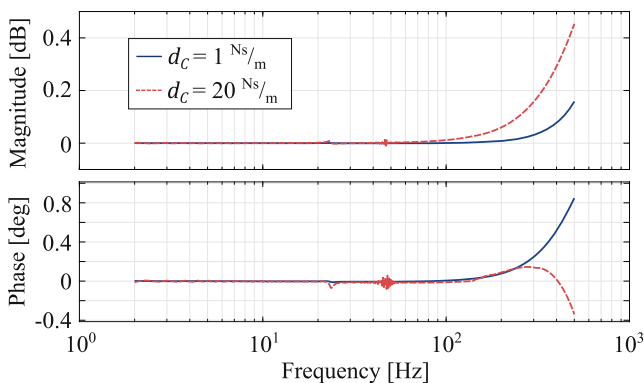


Fig. 11 Simulated frequency response of the closed loop control $F_{C,ref}$ to F_C with ideal measurement of the state variables

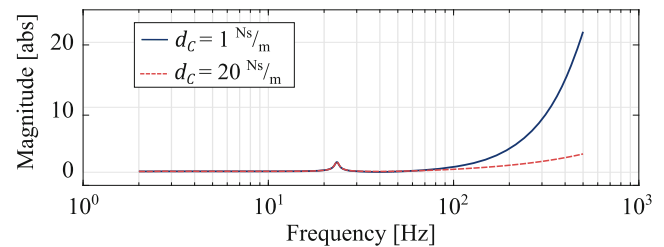


Fig. 12 Simulated current requirement for the force control (transmission behaviour $F_{C,ref}$ to i) with ideal measurement of the state variables

the Kalman filter are chosen such that the model tends to be trusted more than the measured quantities:

$$\begin{aligned}
 \underline{Q}_x &= \text{diag} [5 \cdot 10^{-4} \ 1 \cdot 10^{-6} \ 1 \cdot 10^{-4} \ 2 \cdot 10^{-7}], \\
 \underline{R}_x &= \text{diag} [8 \cdot 10^{12} \ 3 \cdot 10^{10}], \underline{N}_x = \underline{0}.
 \end{aligned}$$

6 Simulation results

Figure 10 shows the schematic structure of the simulation model. The model of the test bench, consisting of power amplifier, shaker and bending beam, corresponds to the surrogate model from Sect. 4. The current control, force control and observer were implemented as described above and simulated with the frequency of the dSPACE board of 15 kHz. The reference signal $F_{C,ref}$ is a sinusoidal sweep whose derivative $\dot{F}_{C,ref}$ is calculated analytically. The following simulative tests were carried out with an amplitude of 1 N with the goal to validate the chosen control concept. Parameters were varied manually to understand the effects of certain mechanical properties contained in the linearisation Eq. 5.

6.1 Ideal measurement of system states

First, the control concept is validated by assuming an ideal measurement of all required state variables. It has to be

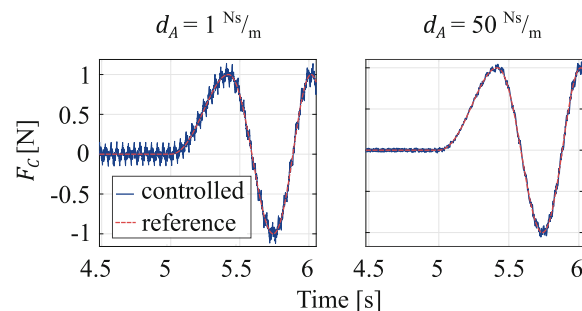
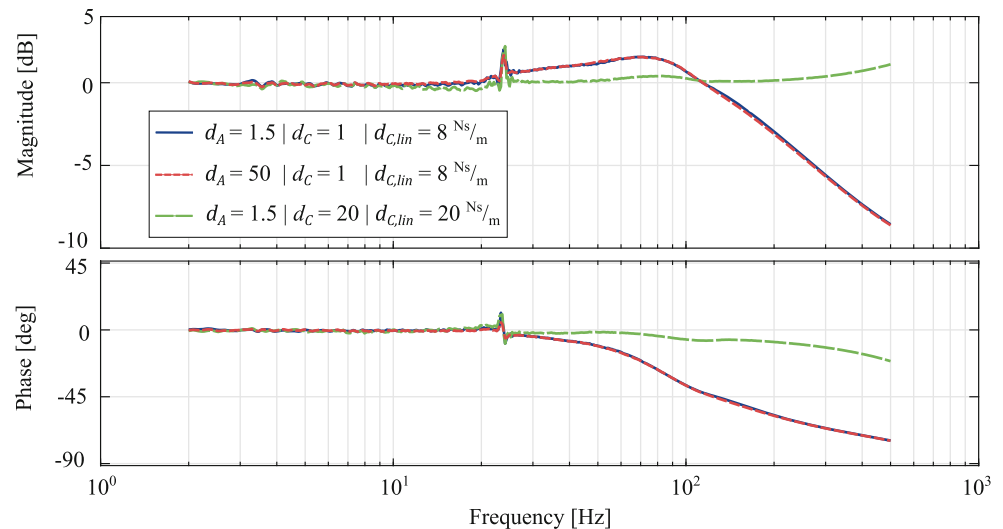


Fig. 13 Left: simulated target-actual curve of force control with ball bearing ($d_A = 1,5 \frac{Ns}{m}$); right: simulated target-actual curve of force control with slide bearing ($d_A = 50 \frac{Ns}{m}$)

Fig. 14 Simulated frequency response of the closed loop control for different combinations of the damping parameters d_A , d_C and $d_{C,lin}$



clarified whether the input-output linearisation is also valid for the surrogate model. The controller gain is chosen simulatively as $K = 80 \cdot 2 \cdot \pi$.

As can be seen, the force control is stable with a -3 dB bandwidth above 500 Hz. The assumptions made are therefore valid. The anti-resonance of the shaker-side force has no effect on the control (Fig. 11).

It is also interesting to consider the parameter d_C , since according to Eq. 5 it is inversely proportional to the required Lorentz force or the current to be set. This becomes clear in Fig. 12, where the frequency response of F_c to current i is shown.

A 20-fold increase in the damping value of the contact spring compared to the test bench parameter leads to a significant reduction in the required current. Since according to the data sheet, the maximum permissible current is 11 A, this is a significant influencing factor for the realisable control bandwidth.

6.2 Consideration of measurement and Kalman filter effects

In the following, the simulation is extended to include measurement effects and the Kalman filter. Since the usable frequency ranges of the sensors are far beyond 1 kHz, only

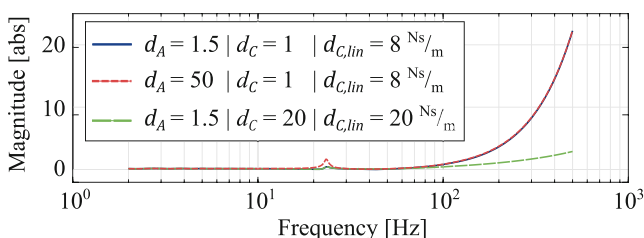


Fig. 15 Simulated current requirement of the force control for different combinations of the damping parameters d_A , d_C und $d_{C,lin}$

the measurement noise was modelled as white noise and supplemented by the quantization of the dSPACE board. All measurement signals also pass through amplitude-dependent nonlinear smoothing filters according to [6] before being fed to the Kalman filter.

Figure 13 shows two curves of the nominal and actual force. The controller gain is still $K = 80 \cdot 2 \cdot \pi$ and the damping of the contact spring $d_C = 1 \frac{Ns}{m}$. To be able to operate the controller at rest without immediate instability, the value for the input-output linearisation is set to $d_{C,lin} = 8 \frac{Ns}{m}$. This reduces the amplification of the measurement noise in the calculation of F_A according to Eq. 5, which can easily lead to a required current beyond limitation.

Two different damping parameters of the armature suspension are simulated with $d_A = 1.5 \frac{Ns}{m}$ and $d_A = 50 \frac{Ns}{m}$. The lower value can be realized on the test bench by replacing the armature's slide bearing with a ball bearing, which should result in lower power consumption according to Eq. 5. However, as can be seen, this results in an oscillation with the natural frequency of the SUT appearing in the control signal, which cannot be removed by the control. By using the slide bearing, this oscillation is considerably reduced.

It can thus be seen that the system damping has a decisive influence on the performance of the control. This can also be derived from Fig. 14, which shows the closed loop transfer functions for different combinations of the damping parameters.

Table 2 Parameters of the damping contact element

Parameter	Description	Value
c_C	Contact element stiffness	7300 N/m
d_C	Contact element damping	23 Ns/m ²
m_{CE}	Contact element mass	45 g

Fig. 16 **a** CAD model of the contact element; **b** test setup with installed contact element

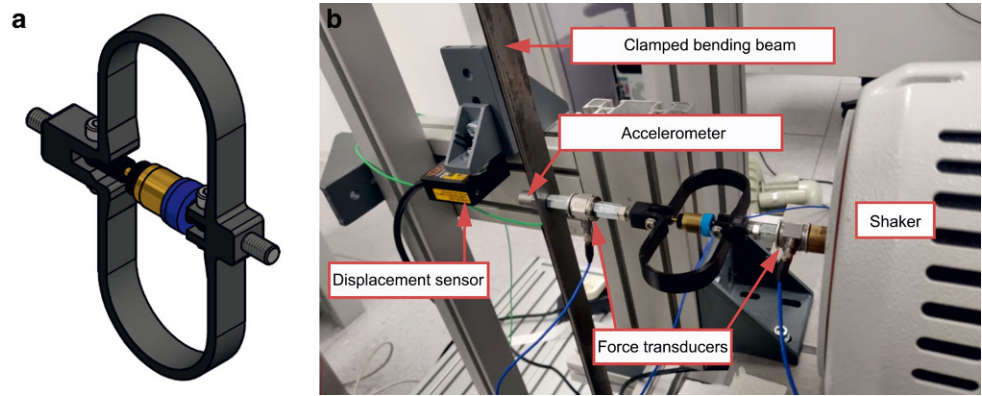
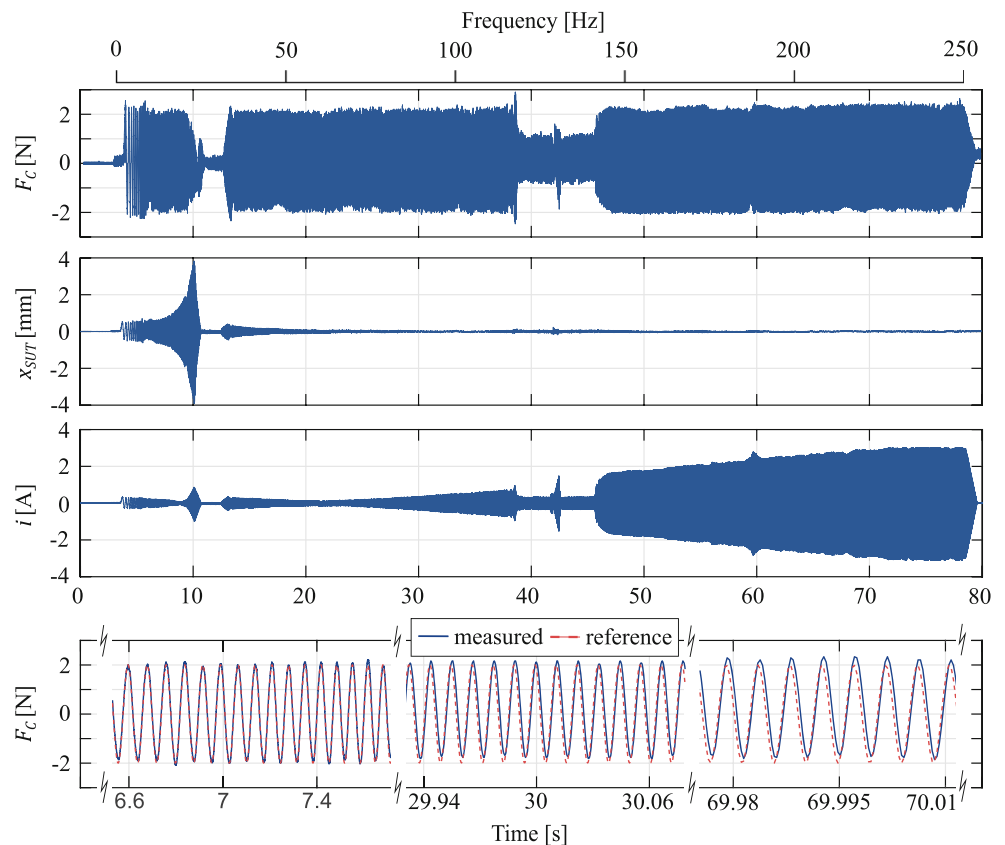


Fig. 17 Controlled contact force, beam deflection, and coil current for a sinusoidal sweep from 2 to 250 Hz on a beam of length 750 mm



For a low contact damping, an increased value for the linearisation must be assumed in order to stabilize the control (blue and red graph). For a higher value of $d_C = 20 \frac{Ns}{m}$, this can also be used in the linearisation, resulting in a higher control bandwidth (green graph). At low frequencies, the force is mainly transmitted via the spring, while at high frequencies the damper force dominates and compensates the low-pass behaviour of the spring. This is also advantageous regarding the current requirement, as Fig. 15 shows. For high frequencies, a high value for d_C is advantageous (green graph).

7 Test bench results

The simulation results lead to the conclusion that the pure spring used as a contact element for the previous tests does not permit high-frequency force control. A new contact element was therefore designed, which comes as close as possible to an ideal spring-damper element and replaces the traditionally used stinger. Figure 16 shows both the CAD model and the component assembled in the test rig. It consists of a plastic spring produced by 3D printing, into which threaded rods are glued on both sides for the application of the force sensors. A compact model shock absorber filled with hydraulic oil is embedded in the centre axis. It contains

Fig. 18 Frequency responses of the closed-loop force control for different setpoint amplitudes on a beam of length 750 mm

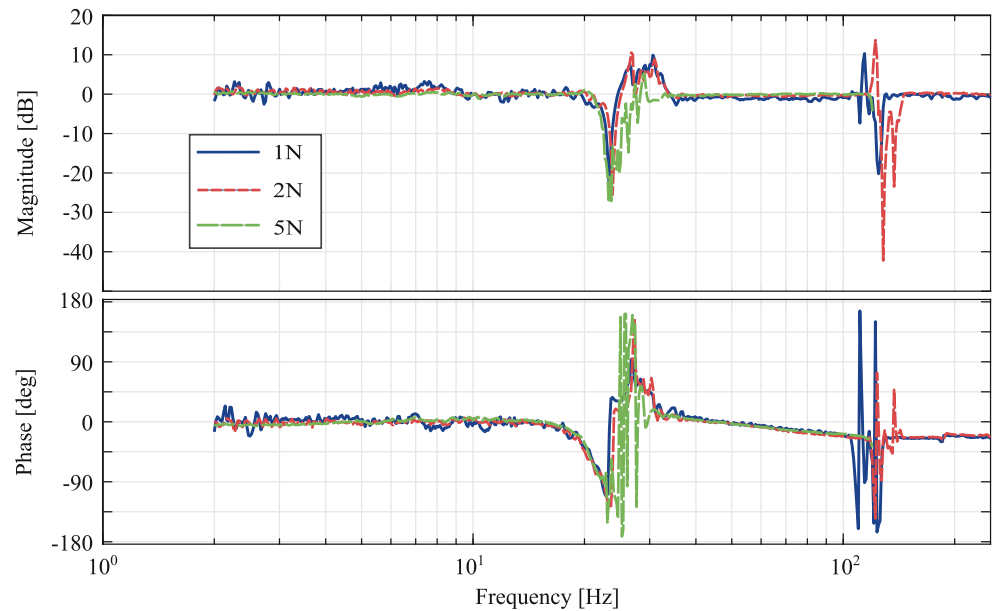
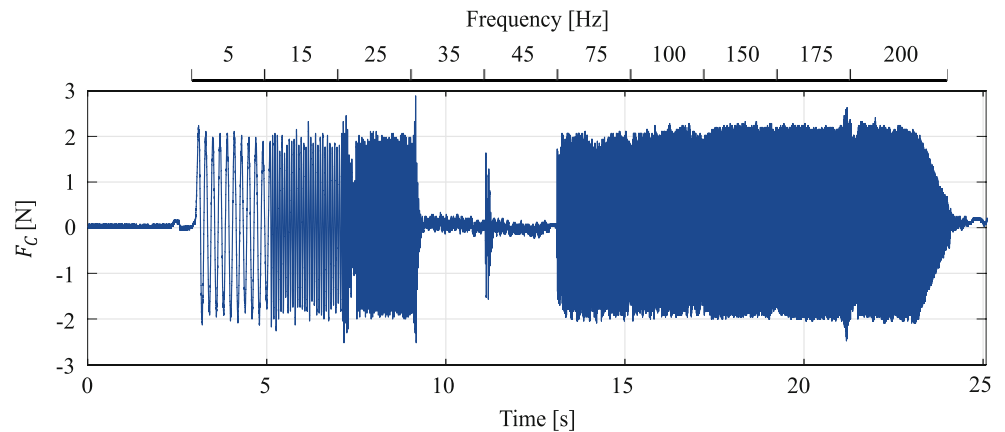


Fig. 19 Force signal of a stepped-sine excitation with nominal amplitude 2 N on a beam of length 550 mm



a piston which generates the damping force by moving in the oil. The substitute parameters for the force control were determined as depicted in Table 2.

To be able to achieve a good control result, the setpoint signal generation was extended by two components, namely a continuous control of the force amplitude and an automatic reduction of the setpoint force around the SUT resonance.

The control of the force amplitude is necessary to compensate for the deviations of the design model from the real test bench behaviour, e.g. by measurement errors, unmodeled physical effects or time varying parameters during testing. In particular, the mass of the contact element and the dynamics of the damper, which are difficult to model, are among these factors. Following the algorithm from [24], the maximum value of the force signal A_{max} is determined over a time interval of 0.1 s. For the next interval, the mod-

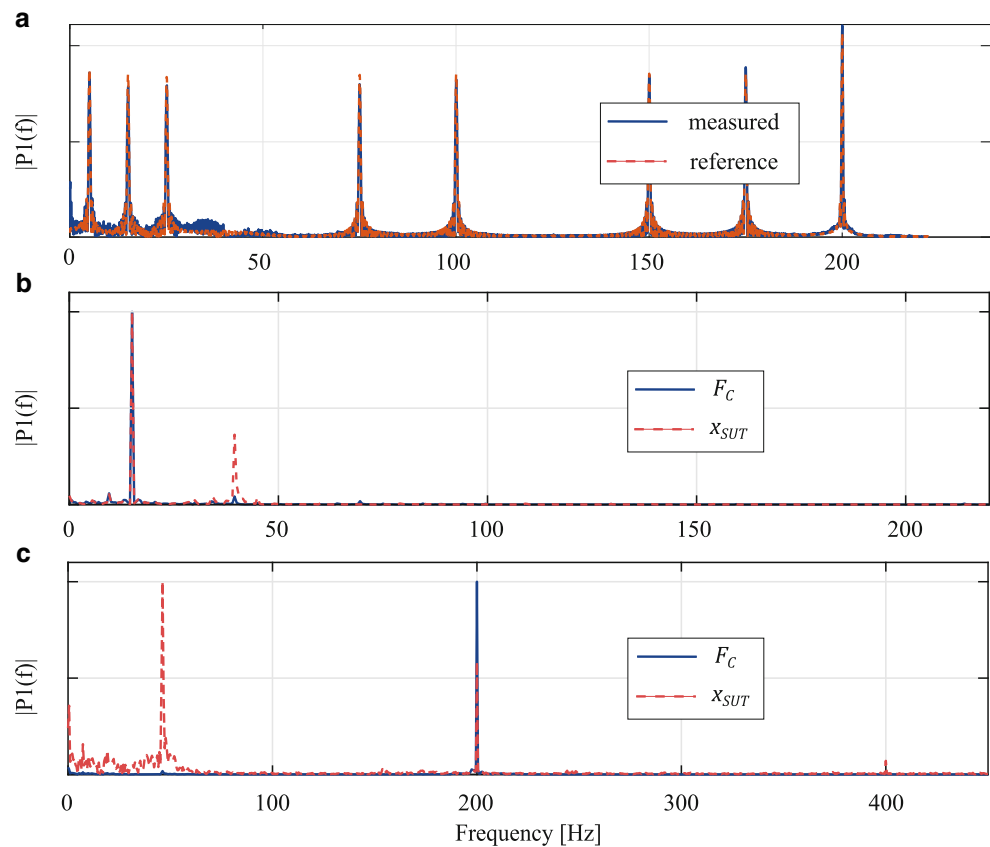
ified amplitude of the nominal signal is corrected by the difference between the nominal and actual amplitude:

$$A_{mod,k} = A_{mod,k-1} + (A_{soll} - A_{max,k-1}). \quad (13)$$

The lowering of the target signal in areas of the SUT resonance is necessary because, on the one hand, damage to the test rig is to be avoided. In addition, due to the aforementioned transfer zero problem, the force dropout phenomenon cannot be completely avoided. Therefore, the measurement signals of the SUT are monitored continuously. As soon as the algorithm detects an extreme increase in the SUT acceleration, the target signal has to be reduced to an amplitude of 0.1 N for a frequency range of 11 Hz in order to be able to pass through the beam resonance regions.

The time curves of the force control can be seen in Fig. 17. The target signal is a sinusoidal sweep with an amplitude of 2 N. Since the model shock absorber is proba-

Fig. 20 **a** reference and measured frequency spectrum of the force signal of a stepped-sine excitation with nominal amplitude 2 N at a beam of length 550 mm; **b, c** frequency spectra of the measured force and the SUT deflection for excitations at 15 Hz and 200 Hz respectively



bly not designed for high frequencies, the target signal was terminated at 250 Hz.

The sampling frequency of the real-time board was 15 kHz. The controller gain had to be manually adjusted to $K = 15 \cdot 2 \cdot \pi$. It should be noted that the actuator damping used for linearisation had to be selected as $d_{A,lin} = 1,5 \frac{Ns}{m}$, which does not correspond to the damping of the plain bearing present on the test rig. Otherwise, the activation of the force control (shown in Fig. 17 at 6 s) would immediately lead to instability, as this parameter amplifies the noisy signal of the armature velocity in Eq. 5.

The curves show that the force signal has the desired amplitude until the end of the excitation. The resonances of the beam at 24 and 126 Hz can be seen in the graph of x_{SUT} and lead to the desired lowering of the nominal force signal. As determined in the simulations, the current requirement increases sharply with increasing frequency.

The same excitation was repeated with target amplitudes of 1 and 5 N. The Bode plots of the corresponding transfer functions can be seen in Fig. 18. Outside the SUT resonance points, there is almost perfect guiding behaviour up to a frequency of 250 Hz. The excitation with 5 N target amplitude could be performed successfully only up to the second resonance point due to safety-related limitations regarding the coil current.

A key requirement for the control is that it should work without model knowledge of the SUT. To test this, the beam was shortened by 26.6% to 550 mm, a configuration which was not tested in simulations before. This shifts the first resonance point from 24 to 42 Hz, while the second one is now outside the excitation spectrum. Figure 19 shows a stepped-sine excitation with frequency steps of 5, 15, 25, 35, 45, 75, 100, 150, 175 and 200 Hz. The length of each step is 1.5 s. Again, the target amplitude is well controlled over wide ranges. Short-term dips or rises after a new frequency step is applied are quickly corrected by the amplitude control. In the range of the beam resonance, the reference signal is lowered as desired, all this without any modifications of the control parameters.

The sinusoidality of the signal is also given. This can be seen from the frequency spectrums in Fig. 20. The spectrum of the controlled force signal essentially matches the desired frequency components of the reference signal. The peak at 200 Hz is higher because the reference signal is ramped to zero at this frequency at the end and the frequency is therefore present for a longer time. Meanwhile, the spectra of the SUT deflection contain further harmonic components that are almost absent in the contact force, which is exemplarily shown for the excitation frequencies 15 Hz and 200 Hz.

The measurements show that the developed control concept is suitable for a highly dynamic and accurate force

control. As expected, the integration of a damping contact element led to largely smooth force curves and a high control bandwidth. The evaluations also suggest that a higher bandwidth than 250 Hz, as presented here, is possible. Concerning the nonlinearity of the SUT, an amplitude dependent resonance shift could not be observed, whereas multiple frequencies in the response signal occur. Despite the mechanical coupling of shaker and SUT, these are not fed back into the force excitation signal thanks to consideration of the SUT behaviour in the linearisation Eq. 5.

8 Conclusion and outlook

In this paper, a novel concept for a control of the contact force in experimental vibration analysis was presented. The main goal was to achieve a high control bandwidth while the dynamics of the test structure is unknown to the control algorithm. The underlying models were derived using measurement data from the real test rig, and the control concept and state observer were presented. The control concept is based on the introduction of a novel contact element, which softens the coupling between the excitation unit and the test structure. The known concept of a compliance spring is enhanced by adding a damping element. The control algorithm is based on input/output linearisation and an underlying current control loop to compensate for nonlinearities of the shaker. An online adaptation of the reference amplitude was implemented to cope with model uncertainties. The algorithm works under total absence of any a priori knowledge of the SUT dynamics, while only requiring real-time measurements of the SUT response, which makes the concept also suitable for other applications such as HIL testing. The validity of the concept as well as the influence of the system parameters on the control performance were investigated simulatively. Based on measurements of the real test rig, it was demonstrated that a bandwidth of the force control of at least 250 Hz is possible, not only controlling the force level, but the complete waveform without time consuming iterative parameter adaptation during the testing process.

Further work could investigate to what extent a more accurate modelling of the contact element could improve the formulation of the control law. In addition, the examination of the SUT resonance behaviour is still limited by input notching. Here, an improvement could be achieved by enhanced resonance point detection and possibly by switching from force to displacement control like suggested in [11]. To emphasise the nonlinear behaviour of the SUT, the beam parameters could be selected so that the resonance is shifted to a lower frequency region.

Funding Open Access funding enabled and organized by Projekt DEAL.

Availability of data and materials Not applicable.

Code availability Not applicable.

Conflict of interest C. Lüke and A. Trächtler declare that they have no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Brandt A (2011) Noise and vibration analysis: Signal analysis and experimental procedures. John Wiley & Sons
2. Cloutier D, Avitabile P, Bono R, Peres M (2009) Shaker/stinger effects on measured frequency response functions. In: Proceedings of the 27th International Modal Analysis Conference, vol 122
3. Chae Y, Rabiee R, Dursun A, Kim CY (2018) Real-time force control for servo-hydraulic actuator systems using adaptive time series compensator and compliance springs. *Earthq Engng Struct Dyn* 47(4):854–871. <https://doi.org/10.1002/eqe.2994>
4. Catalfamo S, Smith SA, Morlock F, Brake MR, Reuß P, Schwing-shackl CW, Zhu WD (2016) Effects of experimental methods on the measurements of a nonlinear structure. In: Dynamics of Coupled Structures, vol 4. Springer, Cham, pp 491–500 https://doi.org/10.1007/978-3-319-29763-7_48
5. Ewins DJ (2009) Modal testing: theory, practice and application. John Wiley & Sons
6. Goot RE, Mahalab U, Cohen R (2005) Nonlinear exponential smoothing (NLES) algorithm for noise filtering and edge preservation. *Hait J Sci Eng* 2(2005):2–10
7. Girard A, Newerla A (2001) Methodology for Notching in Random Vibrations. Fourth International Symposium Environmental Testing for Space Programmes vol 467
8. Hu X (1992) Effects of stinger axial dynamics and mass compensation methods on experimental modal analysis. Diss. Iowa State University
9. Josefsson A, Magnevall M, Ahlin K (2006) Control algorithm for sine excitation on nonlinear systems. In: Society for Experimental Mechanics, IMAC XXIV edn.
10. Kohlstedt A (2020) Model-based synthesis of a hybrid force/position controller for a vehicle suspension test rig with hydraulic hexapod vol 2020. Dissertation, Heinz Nixdorf Institut, Universität Paderborn
11. Karaağaçlı T, Özgüven H (2021) Experimental modal analysis of nonlinear systems by using response-controlled stepped-sine testing. *Mech Syst Signal Process*. <https://doi.org/10.1016/j.ymssp.2020.107023>

12. Lopp GK, Pacini BR, Mayes RL (2009) A Method for Canceling Force Transducer Mass and Inertia Effects. In: Special Topics in Structural Dynamics, vol 5. Springer, Cham, pp 49–60 https://doi.org/10.1007/978-3-319-75390-4_4
13. Mohammadali M, Ahmadian H (2016) Improvement in Modal Testing Measurements by Modeling and Identification of Shaker-Stinger-Structure Interactions. *Exp Tech*. <https://doi.org/10.1007/s40799-016-0009-9>
14. McConnell KG (1995) *Vibration Testing—Theory and Practice*. Wiley
15. McConnell KG, Varoto PS (2008) *Vibration testing: theory and practice*. John Wiley & Sons
16. Noël J, Kerschen G (2016) 10 years of advances in nonlinear system identification in structural dynamics: A review. *Proceedings of ISMA 2016-International Conference on Noise and Vibration Engineering*
17. Niksefat N, Sepehri N (2001) Designing robust force control of hydraulic actuators despite system and environmental uncertainties. *IEEE Control Syst Mag* 21(2):66–77. <https://doi.org/10.1109/37.918266>
18. de Oliveira LPR, Varoto PS, Peres MAS (2011) Shaker structure interaction—Overview and updated results. In: 18th International Congress on Sound and Vibration 2011, ICSV18 edn. Rio de Janeiro, Brazil
19. Phillips AW, Allemang RJ (2003) An overview of MIMO-FRF excitation/averaging/ processing techniques. *J Sound Vib* 262(3): 651–675. [https://doi.org/10.1016/S0022-460X\(03\)00116-0](https://doi.org/10.1016/S0022-460X(03)00116-0)
20. Pratt GA, Williamson MM, Dillworth P, Pratt J, Wright A (1997) Stiffness isn't everything. In *experimental robotics IV*. Springer, Berlin, Heidelberg, pp 253–262 <https://doi.org/10.1007/BFb0035216>
21. Pye C (2016) Accurate Frequency Measurement on Small Structures with Shaker Excitation. *Topics in Modal Analysis & Testing*. In: *Conference Proceedings of the Society for Experimental Mechanics Series*. Springer, vol 10. Cham https://doi.org/10.1007/978-3-319-30249-2_36
22. Robbins E, Kuether RJ, Pacini BR, Moreu F (2023) Stabilizing a strongly nonlinear structure through shaker dynamics in fixed frequency voltage control tests. *Mech Syst Signal Process*. <https://doi.org/10.1016/j.ymssp.2023.110118>
23. Robinson DW (2000) *Design and Analysis of Series Elasticity in Closed-loop Actuator Force Control*. Massachusetts Institute of Technology, Massachusetts (Dissertation)
24. Ricci S, Peeters B, Debille J, Britte L, Faignet E (2008) Virtual Shaker Testing—a Novel Approach for Improving Vibration Test Performance. *International Conference on Noise and Vibration*. ISMA, vol 2008. Engineering, Leuven, pp 1767–1782
25. Scheel M (2022) Nonlinear modal testing of damped structures: Velocity feedback vs. phase resonance. *Mech Syst Signal Process*. <https://doi.org/10.1016/j.ymssp.2021.108305>
26. Snyder D, Lang GF (2001) Understanding the Physics of Electrodynamical Shaker Performance. In: *Sound and Vibration*, vol 4
27. Sivaselvan MV, Reinhorn AM, Shao X, Weinreber S (2008) Dynamic force control with hydraulic actuators using added compliance and displacement compensation. *Earthq Engng Struct Dyn* 2008(37):1785–1800. <https://doi.org/10.1002/eqe.837>
28. Tang B, Brennan MJ, Gatti G (2018) Use of the dynamic stiffness method to interpret experimental data from a nonlinear system. *J Sound Vib* 421:91–110. <https://doi.org/10.1016/j.jsv.2018.01.006>
29. Varoto PS, de Oliveira LPR (2002) On the Force Drop Off Phenomenon in Shaker Testing in Experimental Modal Analysis. *Shock Vib* 9(4–5):165–175
30. Varoto PS, de Oliveira LPR (2002) Interaction Between a Vibration Exciter and the Structure Under Test. *Sound Vib*: 20–25
31. Vukobratović M (1997) How to control robots interacting with dynamic environment. *J Intell Robot Syst* 19(2):119–152. <https://doi.org/10.1023/A:1007974811131>
32. Waimer S, Manzato S, Peeters B, Wagner M, Guillaume P (2018) Modelling and simulation of a closed-loop electrodynamic shaker and test structure model for spacecraft vibration testing. *Adv Aircr Spacecr Sci* 5(2):205–223. <https://doi.org/10.12989/aas.2018.5.2.205>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.